Food Coupon QR Code System: Process Flowchart

This flowchart details the workflow of the Food Coupon QR Code System, outlining the interactions between the Event Admin and the System.

Phase 1: Pre-Event Setup

1. **START: Admin Portal**
    - Admin logs into the web-based admin portal.
2. **Admin Action: Import Participant Data**
    - Admin navigates to the "Import CSV" page.
    - Selects and uploads a CSV file containing: Team Name, Member Name, Email ID.
3. **System Process: Database Population**
    - The system parses the CSV file.
    - For each participant, it creates a record in the database with:
        - A unique, unguessable ID (this will be the QR code data).
        - Participant details (Name, Team, Email).
        - Initial status for all meals (e.g., Meal 1, 2, 3, 4, 5) set to **"Not Eaten" (Red)**.
        - Coupon status set to **"Enabled"**.
4. **Admin Action: Configure Meal Schedule**
    - Admin navigates to the "Schedule" page.
    - Sets the start and end times for each meal (e.g., Breakfast, Lunch, High Tea, Dinner).
    - The system uses this schedule to determine the "current meal" and can automatically enable/disable scanning periods.
5. **Admin Action: Distribute QR Codes**
    - Admin navigates to the "Home" page.
    - Clicks the **"Send QR Codes to All Participants"** button.
6. **System Process: QR Code Generation & Email**
    - The system iterates through every participant in the database.

- It generates a unique QR code image for each participant based on their unique ID.
- *(Simulation)*: An email containing the personal QR code is sent to each participant's registered email address. This QR code is valid for all meals in the event.

Phase 2: During the Event (Meal Time)

1. **START: Meal Service Begins**
   - Participants arrive with their QR codes on their mobile devices.
   - Admin opens the "Scan QR" page on a scanning device (laptop/tablet with a camera).
2. **Admin Action: Scan Participant's QR Code**
   - Admin aims the device camera at the QR code.
3. **System Process: Decode and Validate**
   - The system decodes the QR code to extract the participant's unique ID.
   - It then performs a series of checks against the database:
     - **Is the ID valid?** (Does it exist in the database?)
     - **Is the coupon enabled?** (Has the admin globally disabled coupons?)
     - **Has this meal already been claimed?** (Is the status for the *current meal* already "Eaten"?)
4. **Decision Point: Is the Scan Valid?**
   - **IF YES (All checks pass):**
     - **System Action:** Updates the database.
       - Changes the participant's status for the current meal to **"Eaten" (Green)**.
     - **System Feedback:** Displays a **"SUCCESS"** message with the participant's name and team name on the scanner screen.
     - **System Action:** The "Logs" page and "Dashboard" stats are updated in real-time to reflect the change.
     - **-> END: Meal Served to Participant.**
   - **IF NO (Any check fails):**

        - **System Feedback:** Displays an **"ERROR"** message.
            - "Invalid QR Code" (if ID not found).
            - "Coupon Already Used for this Meal" (if status is "Eaten").
            - "Coupon Currently Disabled" (if admin has disabled it).
        - **-> END: Participant Denied Meal.**

Phase 3: Admin Monitoring & Management

This phase happens concurrently with Phase 2.

1. **Admin Action: View Logs**
    - Admin navigates to the "Logs" page.
    - **System Display:** Shows a list of all teams. Beside each team name, status squares for each member indicate their current meal status (Red, Green, or Gray for disabled). Hovering over a square reveals the member's name.
2. **Admin Action: View Participant Details**
    - Admin clicks on a team name or a specific member's status square in the logs.
    - **System Display:** Navigates to a new page showing:
        - Team Name.
        - Details of the specific member (Name, Email).
        - A button to **"Resend QR Code Email"** to just that member.
3. **Admin Action: Use Global Settings**
    - Admin navigates to the "Settings" page.
    - Admin can click one of three buttons:
        - **RESET:** Changes the status of the current meal for **all** participants back to "Not Eaten".
        - **ENABLE:** Enables coupons for all participants who have not yet eaten the current meal.
        - **DISABLE:** Globally disables all coupons for all participants, preventing any further scans until re-enabled.